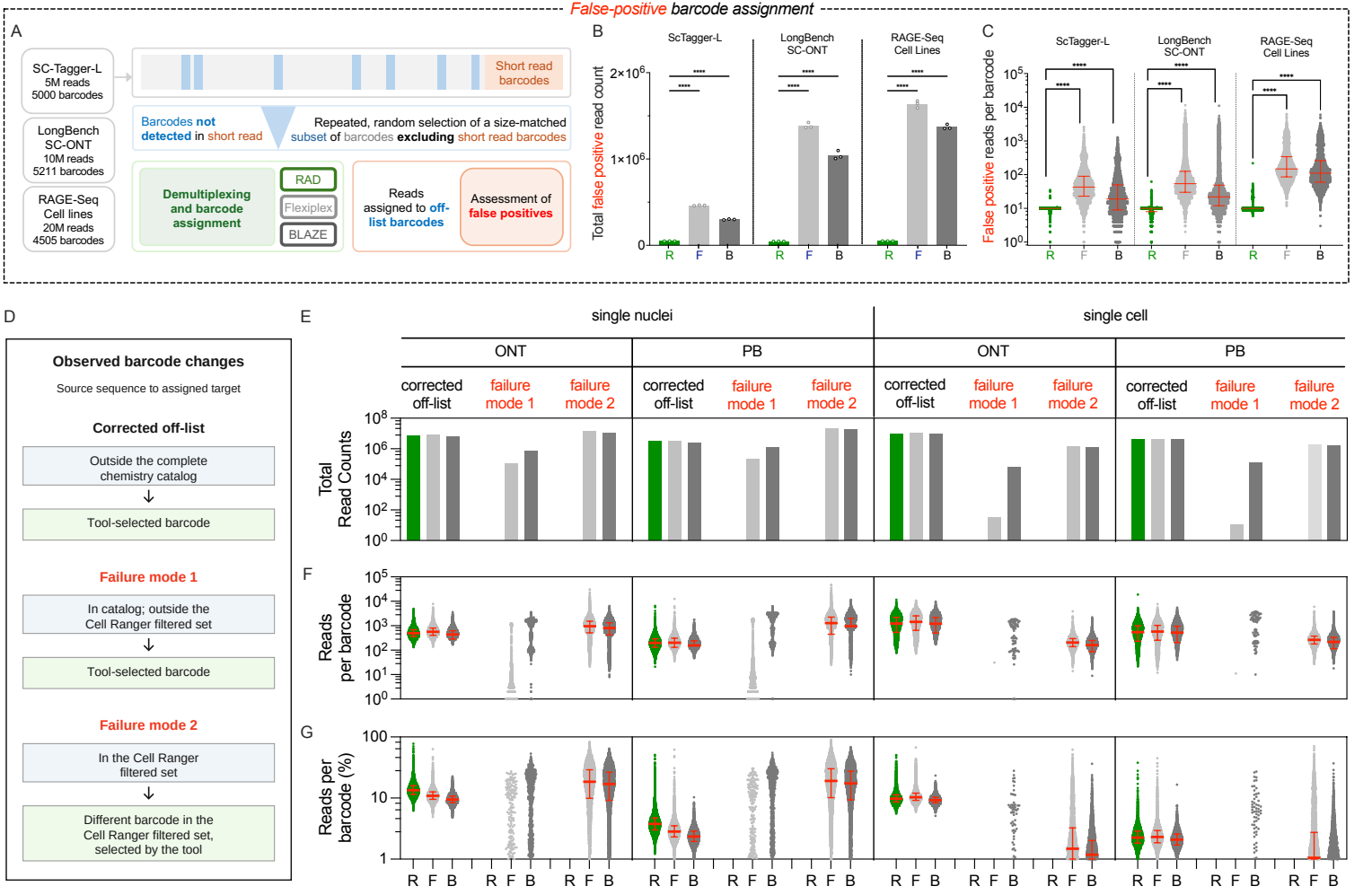
False-positive barcode assignment
A
SC-Tagger-L
5M reads
5000 barcodes
LongBench
SC-ONT
10M reads
5211 barcodes
RAGE-Seq
Cell lines
20M reads
4505 barcodes
Short read barcodes
Barcodes not detected in short read
Repeated, random selection of a size-matched subset of barcodes excluding short read barcodes
Demultiplexing and barcode assignment
RAD
Flexiplex
BLAZE
Reads assigned to off-list barcodes
Assessment of false positives
B
ScTagger-L
LongBench SC-ONT
RAGE-Seq Cell Lines
Total false positive read count
2×10^6
1×10^6
0
****
R F B
C
False positive reads per barcode
10^5
10^4
10^3
10^2
10^1
10^0
D
Observed barcode changes
Source sequence to assigned target
Corrected off-list
Outside the complete chemistry catalog
Tool-selected barcode
Failure mode 1
In catalog; outside the Cell Ranger filtered set
Tool-selected barcode
Failure mode 2
In the Cell Ranger filtered set
Different barcode in the Cell Ranger filtered set, selected by the tool
E
single nuclei
single cell
ONT
PB
corrected off-list
failure mode 1
failure mode 2
Total Read Counts
10^8
10^6
10^4
10^2
10^0
F
Reads per barcode
10^5
10^4
10^3
10^2
10^1
10^0
G
Reads per barcode (%)
100
10
1
R F B